Quiz — 1.2.1 Operating Systems

Name: ___ Date: _ ***Mark:*** ___ / 25

OCR H446 · Component 01 · 1.2.1

Section A — Multiple choice (8 marks)

1. Which statement best describes virtual memory? A. Extra RAM chips added to increase memory capacity B. Using secondary storage as if it were RAM to run more/larger programs C. Cache memory built into the CPU D. A region of ROM reserved for the operating system Your answer: ☐

2. Paging divides memory into blocks that are: A. Variable-sized and based on the program's logical structure B. Fixed-size C. The same size as the page file only D. Always 1 KB Your answer: ☐

3. Which type of fragmentation is associated with segmentation? A. Internal fragmentation B. External fragmentation C. No fragmentation occurs D. Disk fragmentation only Your answer: ☐

4. During interrupt handling, the contents of registers such as the PC and ACC are: A. Deleted to save memory B. Pushed onto a stack so the task can resume C. Written to the page file D. Sent to the printer buffer Your answer: ☐

5. A real-time operating system (RTOS) is defined by its ability to: A. Run on many networked machines as one system B. Guarantee a response within a set time limit C. Allow many users to log in at once D. Fit entirely within ROM Your answer: ☐

6. Which scheduling algorithm gives each process a fixed time slice (quantum) in turn? A. Shortest job first B. First come first served C. Round robin D. Shortest remaining time Your answer: ☐

7. Which is the FIRST program to run when a computer is switched on? A. The operating system kernel B. The BIOS C. The device driver D. The loader Your answer: ☐

8. Java bytecode run by the JVM is an example of: A. A system virtual machine running a whole guest OS B. Intermediate code run by a process virtual machine C. A device driver D. An interrupt service routine Your answer: ☐

Section B — Short answer (12 marks)

9. State **two** differences between paging and segmentation. **[2]**

10. Explain what is meant by *disk thrashing* and why it slows a computer. **[2]**

11. Describe the steps the CPU/OS takes to handle an interrupt, making clear why the stack is needed. **[3]**

12. A washing machine and a mainframe use different types of operating system. Name the most suitable OS type for each and justify your choice. **[2]**

13. Explain the role of a device driver and state why it is both hardware- and OS-specific. **[3]**

Section C — Exam-style question (5 marks)

14. A software company wants to test an unfamiliar piece of software on several different operating systems without buying extra physical computers. Explain how a **virtual machine** could be used to meet this need, and discuss **one** advantage and **one** drawback of this approach. **[5]**

Answer key

Section A (8 marks — 1 each) 1. **B** — virtual memory uses secondary storage (swap/page file) as if it were RAM; it does not add physical RAM. 2. **B** — pages and frames are fixed-size; variable blocks describe segmentation. 3. **B** — variable-size segments leave gaps → external fragmentation (paging → internal). 4. **B** — registers (PC, ACC) are pushed onto a stack so the interrupted task can resume. 5. **B** — an RTOS guarantees a response within a deadline (e.g. ABS, pacemaker). 6. **C** — round robin uses a fixed quantum per process in turn. 7. **B** — the BIOS (firmware on ROM) runs first, performs POST, then loads the OS. 8. **B** — source compiled to platform-independent intermediate code (bytecode) run by a process VM (JVM).

Section B (12 marks)

9. **[2]** Any two (1 each): paging uses **fixed-size** blocks, segmentation **variable-size**; paging divided by OS/physical memory, segmentation by the program's **logical structure**; paging suffers **internal** fragmentation, segmentation **external**.

10. **[2]** Disk thrashing = excessive paging/swapping of pages between RAM and disk (1); so much time is spent moving pages that little useful processing happens and the machine slows to a crawl (1).

11. **[3]** Any three: CPU checks the interrupt register at the **end of each FDE cycle** (1); finishes the current instruction / compares priorities (1); **pushes the registers (PC, ACC) onto the stack** (1); loads the address of and runs the correct **ISR** (1); **pops registers off the stack** to resume the task (1) — stack is **LIFO** so nested interrupts resume in the correct order (1). Max 3.

12. **[2]** Washing machine → **embedded OS** (one dedicated task, limited resources, stored in ROM, reliable/efficient) (1). Mainframe → **multi-user OS** (many users share resources via scheduler + permissions) (1).

13. **[3]** A device driver is software that lets the OS control/communicate with a specific hardware device (1), translating OS commands into instructions the device understands (1); it is hardware-specific because each device has its own commands and OS-specific because it must use that OS's interfaces/conventions (1).

Section C (5 marks) 14. **[5]** Up to 5 from: a **virtual machine** is software that emulates a complete computer (1); a **system VM** can run a whole guest OS inside the host (1), so the company can install several guest OSes on one physical machine and test the software on each (1). Advantage: saves cost / no extra hardware needed, and provides a safe **sandbox** so untrusted software cannot harm the host (1). Drawback: the VM runs **slower than native** hardware / the VM consumes host resources (RAM, CPU) (1).

Total: 8 + 12 + 5 = 25